Rik L.E.M. Ubaghs

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Bio

Biomedical engineer and founder with expertise in neurotechnology, optical systems, medical devices, and data analytics. I specialize in developing and translating advanced imaging technologies into practical clinical applications, connecting cutting-edge research with meaningful medical impact.

Education

ETH Zurich, Zurich, Switzerland July 2017 - May 2023
Ph.D. Neurotechnology

ETH Zurich, Zurich, Switzerland Mar 2025 - Mar 2026
CAS Entrepreneurial Leadership in Technology Ventures

University of Amsterdam, Amsterdam, the Netherlands Sept 2014 - June 2016
Research Master Brain and Cognitive Sciences, Cum Laude

Maastricht University, Maastricht, the Netherlands Sept 2011 - June 2014
Bachelor Psychology and Neuroscience, Cum Laude

Skills

Languages:	Dutch (Native), English (Fluent), German (Basic)
Programming:	Python, Rust, Matlab, R
Databases/Querying:	SQL, MySQL
Skills:	Data Science, Biomedical Imaging, Optical Systems, Optical design (Zemax, Optiland) Mechanical Design (Autodesk, Solidworks)

Industry Positions

Arago Labs June 2023 - Current
Co-founder, CEO

- Founded an early-stage medical technology venture focused on improving surgical precision in oncology.
- Designed advanced optical imaging platform combining nonlinear microscopy, and ML/DL-based tissue classification.
- Defined market entry strategy, pricing model, reimbursement pathways, and go-to-market roadmap.

Agora Technology, Maastricht, The Netherlands Nov 2016 - Feb 2018
Founder

- Built data solutions that aimed at streamlining/automating the workflow of medical facilities, especially focusing on surgical equipment.
- We focused on internal logistics, and the creation, maintenance, and analysis of large-scale medical equipment databases using ML/DL.

Research Positions

Inst. of Neuroinformatics, ETH Zurich, Zurich, Switzerland July 2017 – June 2023
Ph.D. Candidate

- Development of a MRI compatible florescence microscope (full-cycle product design).
- Design of a paradigm to combine microscopy and MR imaging methods in awake rodents.
- Design of data pipelines for multi-modal data analysis.

Center for Neuroengineering, Duke University, Durham, USA Dec 2015 – Nov 2016
Visiting Scientist

- Design of a hybrid brain decoding strategy in Non-human Primates.
- Development of ML/DL decoding pipeline for brain decoding.

Selected publications and patents

Publications:

- **Ubaghs, R.L.E.M.**, … Grewe, B.F. Simultaneous single-cell calcium imaging of neuronal population activity and brain-wide BOLD fMRI. *Nature Methods* (in press).
- Rychen, J., … **Ubaghs, R.L.E.M.**, … Hahnloser, R. Recordings of cetacean vocal behaviour with hydrophone arrays in Northern Norway. *bioRxiv* (2023); under review *Nature Scientific Data*. https://whalesonic.stream.

Patents:

- Sainz Villalba, L., **Ubaghs, R.L.E.M.** Device and method for providing histological navigation information during in-vivo surgery. European Patent Application, Filed 2026.

Grants and fellowships

- ETH Zurich Pioneer Fellowship April 2025 - Current
- Swiss Cancer Foundation research grant September 2026
- Galbiati Foundation research donation June 2025